

The Industrial Stack Behind Artificial Intelligence

Where Economic Rent Survives Across the AI Infrastructure Chain, 2026–2036

Data current to Q2 2026. Numbers tagged by source tier (T1 filings/audited; T2 recognized industry research; T3 estimates and analytical judgement) and confidence. Geography is treated as manufacturing concentration and supply-chain risk only.

1. Executive conclusion

AI has become a physical industrial system, and its binding constraint has moved down the chain into power, grid, and memory. Constraint location does not determine rent location: a physical bottleneck creates durable rent only when scarcity cannot be relieved through capital, competition, substitution, or incumbent absorption. The principle is applied below and not repeated.

Durable rent stays concentrated in EUV lithography, the leading-edge foundry, electronic design automation, and the leading accelerator ecosystem. A middle group — high-bandwidth memory, grid equipment, the qualified power-device tier — earns the temporary returns of a capacity cycle and ranks low once economics are taken on a normalized rather than peak basis. Most energy and physical-infrastructure layers are strategically necessary but economically competitive. The battery and storage leader is, on current evidence, the physical-layer exception. The aim is not to identify the most strategically important participants but to identify where rent can survive competition over a decade.

2. Evidence architecture

Each core conclusion as claim, mechanism, evidence with tier, confidence, and falsification condition. Confidence is highest for current structure and lowest for future rent migration.

C1 — Constraint is not rent. Mechanism: capital relieves reproducible scarcity, so rent survives only where the barrier rises as capital arrives. Evidence: power and interconnection are the current binding constraints (T2, medium); generation and grid hardware are reproducible and earn the cost of capital (T1 utility economics, high). Confidence: high on mechanism. Falsification: a physical layer develops a proprietary standard capital cannot replicate.

C2 — Rent concentrates upstream. Mechanism: escalating capital and tacit yield-learning raise the leading-edge barrier faster than entrants can climb it; the critical lithography step has no commercial alternative. Evidence: commercial EUV lithography remains supplied exclusively by one firm (T3, high); the leading foundry holds roughly two-thirds to seventy percent of pure-play foundry revenue, up from about 64% in 2024 (T2 TrendForce); the design-tool trio holds roughly three-quarters of its market (T2/T3). Confidence: high. Falsification: AI value shifts downstream and chokepoints become cost-plus suppliers.

C3 — Power and physical infrastructure commoditize. Mechanism: reproducible capacity competes returns toward cost even under acute demand. Evidence: dominant global solar capacity with sector losses (T1 filings, high); a leading wide-bandgap substrate specialist insolvent in mid-2025 after roughly a 40% price decline (T2, high). Confidence: high on pattern, medium on duration. Falsification: a proprietary power-delivery standard with lock-in forms.

C4 — China shows high capability and divergent returns. Mechanism: a system optimizing for resilience and capability does not optimize for return on capital. Evidence: dominant solar capacity with losses, a subsidized and sub-economic leading edge, one profitable physical-layer participant in batteries (T1/T2). Confidence: high on the divergence. Falsification: domestic frontier ecosystems become self-sustaining and economically competitive.

C5 — Accelerator rent is durable but less certain than equipment monopolies. Mechanism: software lock-in raises switching costs, but architecture markets evolve faster than equipment markets. Evidence: very high current gross margin and an intact installed base (T1, high); growing hyperscaler custom silicon (T2/T3, medium). Confidence: high on current economics, lower on duration. Falsification: abstraction layers and captive silicon migrate inference at scale.

3. The assessment framework

Economic Rent Assessment = Structural Capability × Economic Capture × Duration.

This is a calibrated decision framework that converts qualitative economic characteristics into comparable assessments, not a scientific prediction model. Each component is a weighted 1-10 composite; the product is scaled to 0-10.

Multiplication is used because durable rent requires all three conditions simultaneously. A participant with high margins but weak durability, high concentration but weak economics, or strong technology but no pricing power does not qualify as durable rent. Addition would let one strong dimension offset a disqualifying weakness; multiplication does not.

What the framework measures: the durability of economic rent, replication barriers, pricing-power persistence, and competitive structure.

What the framework does not measure: stock valuation, investment timing, management quality, geopolitical probability, and short-term earnings. These require separate analysis and are not inputs to the scores.

Component weights.

Structural Capability	w	Economic Capture	w	Duration	w
Market concentration	25%	ROIC	30%	Replacement difficulty	25%
Replication difficulty	25%	Gross margin	20%	Switching costs	25%
Technology moat	20%	Operating margin	20%	Technology-transition resistance	20%
Qualification barriers	15%	Pricing power	20%	Customer-dependence resistance	15%
Ecosystem dependence	15%	Cash conversion	10%	Capital barrier	15%

Economic Capture is scored on **normalized** (through-cycle) economics for cyclical layers, not peak, per Section 5.

Layer	Struct /10	Econ /10	Duration /10	Assessment /10	Archetype
EUV lithography	9.9	9.0	9.1	8.1	Structural monopoly
Leading-edge foundry	8.6	9.0	7.7	6.0	Durable oligopoly
Design tools (EDA)	8.2	8.9	7.9	5.8	Durable oligopoly
Accelerator + software	8.3	9.7	6.4	5.2	Durable oligopoly (valuation caveat)
Semiconductor equipment	6.6	7.6	6.4	3.2	Durable oligopoly (lower)
Battery/storage leader	6.6	6.2	6.0	2.5	Current rent exception
Advanced packaging	6.9	7.0	5.0	2.4	Bottleneck, incumbent-held
HBM memory	7.0	4.3 (norm.)	5.0	1.5	Cyclical pricing power
Grid / qualified power semis	5.7	4.0 (norm.)	4.5	1.0	Cyclical pricing power
HVDC / optical interconnect	5.3	3.0	4.0	0.6	Ownership uncertain
Commodity energy / devices	2.7	1.7	4.0	0.2	Scale without rent

The ordering is framework-driven: one layer clears all three components; the durable oligopolies cluster in the middle; cyclical layers rank low once economics are normalized.

4. Semiconductor analysis

Each number is retained because it changes the assessment; where measurement varies by definition, ranges are used.

ASML. Source of rent: exclusive supply of the critical lithography step. Replication barrier: a multi-decade supplier ecosystem with no commercial alternative. Customer dependence: total; no substitute. Substitution risk: low. Incumbent absorption risk: none — adjacent to no one. Current economics: gross margin around the low-50s%, operating margin low-30s%, ROIC around the mid-20s% (T1, FY2025). Normalized economics: similar to current, given low cyclicalities and a multi-year backlog. Duration 9.1. Not higher because served-market concentration and cross-border policy can shrink demand; not lower because the barrier is rising. Evidence confidence: high. Falsification: an alternative lithography reaches commercial scale.

TSMC. Source: demand aggregation funding a yield-learning flywheel an integrated rival cannot run. Replication: high, the knowledge tacit. Customer dependence: high. Substitution: low. Absorption: low. Current economics: gross margin around 60%, operating margin around 51%, ROE around 35% (T1, FY2025). Normalized economics: gross margin in the low-to-mid 50s% mid-cycle — still exceptional. Duration 7.7. Not higher because leading-edge capacity is geographically concentrated and the node race resets each generation; not lower because no competitor matches the flywheel at scale. Confidence: high. Falsification: customer-driven second-sourcing fragments the demand that funds the flywheel.

NVIDIA. Source: developer lock-in, proprietary networking, full-stack co-design. Replication: silicon partial, software not yet. Customer dependence: high but concentrated. Substitution: moderate — custom ASICs and abstraction layers. Absorption: moderate — captive silicon at the largest buyers. Current economics: gross margin around 75%, ROIC around 75% (T1). Normalized economics: current ROIC reflects exceptional AI-cycle economics above the historical mid-50s-to-mid-60s% gross-margin range; durability depends on whether software lock-in and ecosystem effects preserve margins after competition increases. Duration 6.4. Not higher because architecture cycles move faster than equipment cycles and buyer concentration creates pressure; not lower because migration remains costly and the installed base is intact. Confidence: high on current economics, lower on duration. Falsification: abstraction plus captive silicon migrate inference at scale.

EDA. Source: mid-design workflow lock-in across a concentrated trio (roughly three-quarters combined share, T2/T3). Replication: high. Customer dependence: high. Substitution: low; no full-flow alternative. Absorption: low. Current economics: gross margin around 80%+ (T1). Normalized economics: similar, on recurring revenue and low cyclicalities. Duration 7.9. Not higher because it is an oligopoly; not lower because the flow itself is the lock-in — changing one tool forces re-running the chain. Confidence: high. Falsification: an integrated or open flow reaches advanced-node parity.

HBM. Source: bandwidth scarcity on a three-supplier oligopoly (shares vary materially by quarter and source). Replication: moderate; three credible suppliers. Customer

dependence: high near-term. Substitution: low near-term. Absorption: low. Current economics: peak-cycle margins reported near 60% (T3). Normalized economics: materially lower — memory has repeatedly fallen to losses when capacity outran demand, which is why the assessment uses normalized rather than peak. Duration 5.0. Not higher because cyclicalities are structural; not lower because the current bandwidth constraint and base-die differentiation are real. Confidence: high on cyclicalities. Falsification: structural capacity discipline holds margins through a downcycle.

Advanced packaging. Source: capacity scarcity, held by the leading foundry. Replication: high. Customer dependence: high. Substitution: low. Absorption: high — the foundry internalizes the value. Current economics: healthy; Normalized: similar but captured. Duration 5.0. Not higher because of the absorption; not lower because the process is hard and capacity-constrained now. Confidence: medium. Falsification: an independent packaging specialist establishes a durable standard.

5. Power, energy, and battery

AI electricity demand is real — heading toward roughly 945 TWh by 2030 (T2 IEA) — but demand scarcity alone does not create durable shareholder rent, because the capacity that meets it is reproducible. Grid, transformers, HVDC, power semiconductors, and storage are evaluated on replication barrier, capital cycle, qualification, and incumbent capture, and most fail the durability test: transformer lead times of roughly four years confer real but cyclical pricing power (T2/T3); the qualified silicon-carbide tier holds brief oligopoly rent that erodes as capacity scales, evidenced by a substrate specialist's mid-2025 insolvency after roughly a 40% price decline (T2). Value flows to existing holders — the leading accelerator franchise extending into power delivery, established equipment suppliers in transformers, and utilities in interconnection.

The battery and storage leader earns roughly 17% net and 27% stationary-storage gross margin (T1, FY2025) because that segment is concentrated, differentiated, and qualification-gated. Current evidence supports a rent exception here; this is not described as permanent, since sodium-ion and continued cost-curve decline could lower the field's floor and erode it.

6. China

China operates a hybrid industrial model in which market competition and national strategic objectives jointly influence capital allocation. Three lenses separate the effects.

Industrial capability: strong in manufacturing scale, batteries, deployment, and a mature semiconductor ecosystem. **Economic return:** largely the inverse — scale

leadership does not confer pricing power, so solar generates losses, the leading edge is sub-economic, and energy infrastructure earns commodity returns. **Strategic objective:** consistent and high; Five-Year Plans evidence industrial priorities — supply security, technological independence — not economic success.

Precision on the AI position requires distinguishing three things. **Deployment:** strong — domestic mature and power silicon plus abundant energy support large-scale inference and industrial AI. **Frontier training:** constrained — restricted advanced lithography, a domestic HBM ecosystem that does not yet reach frontier bandwidth, and leading-edge yield gaps cap frontier-training-class compute. **Export competitiveness:** mixed — globally competitive in batteries, solar, and mature chips; not in leading-edge logic, HBM, or advanced equipment. A system can achieve strategic capability without every sector generating high ROIC; batteries are the exception where structure converts capability into rent.

7. Capital-cycle analysis

Capital inflow is not economic rent. When capacity can be reproduced, competition transfers value from producers to customers. Solar attracted enormous capital and global dominance and produced losses; memory attracts cyclical capital and earns cyclical returns, with DRAM contract prices forecast to rise sharply across 2025-2027 (T3) before the cycle turns; batteries attracted capital across a fragmented field that mostly earns thin margins, the leader excepted; AI infrastructure is drawing the largest current inflows — roughly \$700bn of 2026 operator capital expenditure (T1 guidance aggregated) — into the most reproducible layers. Durable returns accrue to standards, ecosystems, qualification barriers, proprietary knowledge, and irreplaceable tools, not to capital recipients as such.

8. Rent migration map

Historical: electricity rent sat in equipment and standards while generation commoditized; internet rent sat in platforms while transport commoditized; semiconductor rent concentrated at the tool, foundry, and design layers while mature chips commoditized. Current rent: EUV, foundry, design tools, and the accelerator ecosystem. Potential future migration, each with current probability, evidence required, and the condition that would invalidate the current upstream-concentration thesis:

- **AI software-abstraction layer** — current probability low-to-moderate; evidence required: hardware-agnostic inference deployments displacing the incumbent

stack at scale; invalidates the thesis if rent shifts from the accelerator franchise to the abstraction layer.

- **New computing architectures** — current probability low; evidence required: a non-GPU architecture reaching production economics; invalidates if compute rent disperses.
 - **Proprietary energy systems** — current probability low; evidence required: a power-delivery standard consolidating with qualification barriers; invalidates the power-commoditization conclusion specifically.
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9. Falsification and monitoring

Accelerator ecosystem. Failure condition: abstraction layers and custom ASICs migrate inference at scale. Observable indicators: custom-ASIC adoption and capital allocated to it; abstraction-layer development and the hardware-agnostic share of new deployments; customer concentration; margin normalization. Reclassification trigger: sustained margin compression plus rising ASIC share plus falling customer dependence → move from durable oligopoly toward contested.

China frontier. Failure condition: domestic equipment, design tools, HBM, and yield-learning become self-sustaining. Observable indicators: domestic HBM output and frontier bandwidth; EDA adoption at advanced nodes; lithography progress; leading-edge yield economics versus subsidized status; equipment localization. Reclassification trigger: domestic frontier HBM plus improving yield economics plus full-flow domestic tools → the frontier-constraint thesis weakens and a second pole forms.

Power. Failure condition: a proprietary power architecture creates lock-in. Observable indicators: proprietary standards; qualification-barrier formation; depth of incumbent vertical integration. Reclassification trigger: standard consolidation plus qualification barriers plus sustained (not normalizing) pricing → move HVDC and power delivery from MONITOR toward OWN.

10. Scenario framework and final classification

Base case — dominant probability (roughly 50–60%): rent remains upstream. Trigger: continued lithography, foundry, and HBM scarcity; accelerator margins hold; power stays reproducible; China stays frontier-constrained. Winners: EUV, foundry, design tools, accelerator franchise, HBM cyclically, battery leader. Losers: commodity energy and manufacturing, independent power-infrastructure entrants. Evidence required to confirm: stable structure across the monitoring indicators.

Upside case (roughly 20–30%): rent migrates into new layers. Trigger: a power-delivery or interconnect standard consolidates with qualification barriers, or system integration captures downstream value. Winners: the standard-holder, power-delivery and interconnect leaders, integrators. Losers: pure component chokepoints if value shifts downstream; grid commoditizers if power standardizes. Evidence required: a forming standard with switching costs.

Downside case (roughly 15–25%): chokepoints weaken. Trigger: CUDA dependence erodes through ASIC migration; demand normalization relieves scarcity; alternative lithography reaches scale; a China frontier breakthrough forms a second pole; or AI monetization lags capital expenditure. Winners: hyperscaler captive silicon, abstraction-layer owners, compute buyers, cyclical names bought at trough. Losers: the accelerator rent position, the upstream-concentration thesis, anyone who capitalized cyclical scarcity as durable. Evidence required: sustained margin compression at the chokepoints.

Evidence quality. Highest for current structure (T1 financials, existing monopolies and oligopolies); medium for institutional estimates (power demand, market shares); lowest for future rent migration (duration scores, scenario probabilities). The structural classifications are firmer than the duration assessments.

Final classification — a rent-durability classification, not a stock recommendation; security selection requires a separate valuation and expectations overlay. **OWN:** EUV lithography, the leading-edge foundry, design tools, the equipment oligopoly, the accelerator franchise (lower duration certainty, valuation caveat); the battery leader as the current exception. **TRADE:** HBM, grid equipment, the qualified power-device tier — to the capacity-catch-up curve. **MONITOR:** optical interconnect, silicon photonics, advanced packaging, HVDC, AI energy management — against the Section 9 indicators. **AVOID:** most energy infrastructure and commodity manufacturing — solar, generic cells, commodity substrate, standalone cooling; and infrastructure credit priced as though collateral does not decay.

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